Stroke Prediction Project

Overview

The selected dataset is about Stroke, this dataset is used to predict whether a patient is likely t o get stroke based on the input parameters like gender, age, various diseases, and smoking sta tus. The data consists of 5110 rows and each row in the data provides relevant information ab out the patient.

Attributes:

1) id, 2) gender, 3) age, 4) hypertension, 5) heart disease, 6) ever married, 7) work type, 8) Residence type, 9) glucose level, 10) BMI, 11) smoking status, 12) stroke.

Problem:

Stroke is like a heart attack, except it occurs in the blood vessels of the brain. Clots can form in the brain's blood vessels, in blood vessels leading to the brain, or even in blood vessels elsewhere in the body and then travel to the brain. These clots block blood flow to the brain's cells. There is conflict between heart attack and stroke.

Motivation:

Our objective is to analyze the relation between attributes and Stroke and build Machine learning model to detect stroke cases.

The Machine learning Algorithms

We selected two machine learning modes:

- **Logistic regression:** is a supervised learning method for classifying data into discrete outcomes. The logistic regression model passes the outcome of a linear function of features through a sigmoid function to calculate the probability of an occurrence. The model then maps the probability to binary outcomes.

- **Random forest:** is a supervised learning algorithm. The "forest" it builds, is an ensemble of decision trees, usually trained with the “bagging” method. The general idea of the bagging method is that a combination of learning models increases the overall result.

The built/produced Model

Cleaning and pre-processing steps:

We started by checking null values and BMI column have 201 empty values, so we filled the null values with median value of BMI. Also, we deleted unneeded column (id). Machine learning models can't deal with text, so we used Label Encoder that converts classes columns to numbers, every class have certain number. Because, the dataset is unbalance dataset, we used Smote to oversample the minority class (Apply SMOTE to training data only).

About the data splitting:

We split the data into 25% test set and 75% training set.

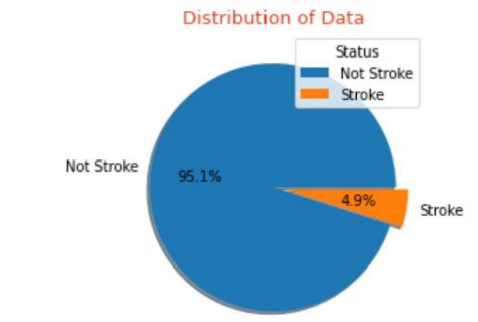
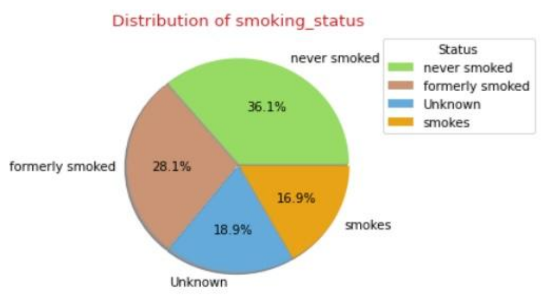
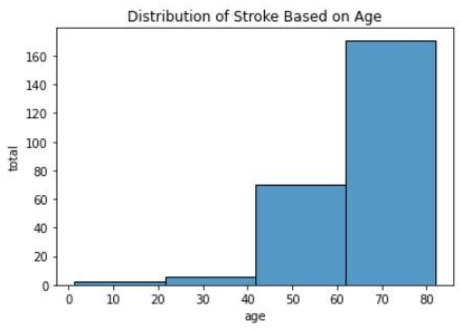
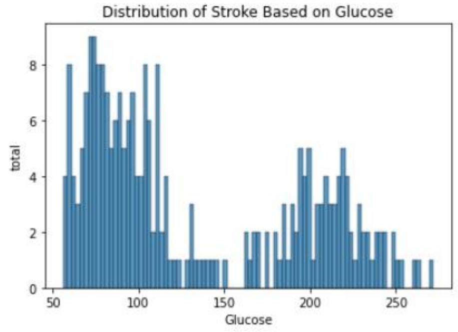
Implementation: (Building the predictive model (Steps and Procedures)):

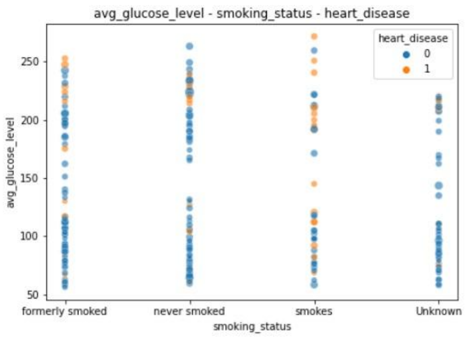
Now the data is cleaned, pre-processed and spitted so, we can train the selected models (Logistic regression, Random Forest) on training data, then used test data in prediction and getting accuracy, and compare the accuracy of both models. because accuracy is not sufficient, we used confusion matrix to know the values of true positive, true negative, false positive and false negative.

The Result Discussion:

This imbalanced dataset has 4.9% of data in the stroke class, so in data splitting, we made sure that the training and testing data included the stroke class. After training the models, the accuracy of Logistic Regression is 76%, and the accuracy of Random Forest is 89%.

The higher accuracy of the Random Forest model suggests its ability to capture complex, non linear relationships in the data, making it more effective in handling the imbalance compared to Logistic Regression, which tends to struggle with minority classes in imbalanced datasets.

	This plot illustrate that the data is imbalance where most of data are non-stroke and about 4.9% stroke
	This plot shows the Distribution of smoking status for stroke data , and it seems that smoking status doesn't tell us anything confounding about stroke-positive individuals, but it may affect with other attributes.
	Higher the Age, Higher the chances of having Stroke
	Higher and lower the Glucose will cause to Higher the chances of having Stroke, so the average value of Glucose should be around 150



The plot illustrates that if persons have normal average value of Glucose and never smokes they will be negative stroke.